Evaluating and Improving Explainability in Facial Emotion Recognition

XAI Final Project

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Dataset: RAF-DB | Model: CNN trained from scratch | XAI: Grad-CAM, LIME, SHAP

Presentation outline

1. Why this problem matters

- Facial Emotion Recognition (FER)
- Trust and transparency requirements
- Research question

2. How we study it

- CNN trained without transfer learning
- Grad-CAM, LIME, SHAP
- Faithfulness, robustness, masking

3. What the results show

- Classification performance
- Correct and incorrect cases
- Quantitative XAI comparison

4. Conclusion

- Main takeaways
- Limitations
- Future work

Motivation: prediction is not enough

Context

- FER classifies face images into 7 emotions: neutral, happy, sad, surprise, fear, disgust, anger.
- CNNs can predict emotions, but their decision process is hard to inspect.
- A correct prediction may still rely on the wrong evidence: background, lighting, framing, or artifacts.

XAI problem

A visually appealing heatmap is not necessarily a faithful explanation of the model.

Central question

Are explanations produced by Grad-CAM, LIME and SHAP truly faithful to model decisions in FER?

Positioning within related work

Peer-reviewed source	Role in our project
Li et al., 2017	RAF-DB: real-world benchmark for facial expression recognition.
Selvaraju et al., 2017	Grad-CAM: gradient-based visual localization for CNNs.
Ribeiro et al., 2016	LIME: local interpretable approximation through perturbations.
Lundberg and Lee, 2017	SHAP: attribution based on Shapley-value principles.
Adebayo et al., 2018	Sanity checks: saliency maps can be misleading.
Yeh et al., 2019	Quantitative evaluation through infidelity and sensitivity.

Our angle

Compare XAI methods on the same FER model using both qualitative and quantitative criteria.

Data and experimental protocol

Dataset

- RAF-DB, basic expression subset.
- 7 emotion classes.
- Aligned face images resized to **128 x 128**, grayscale.
- Fixed train / validation / test splits.

Split sizes

Train	25,120
Validation	5,383
Test	5,384

Reproducibility

- YAML-driven configuration.
- Fixed global seed: 42.
- Final training: 150 epochs; best checkpoint at epoch 135.
- Evaluation on the held-out test split.

Important

The goal is not to beat FER state of the art, but to analyze explanation reliability.

Model: CNN trained from scratch

Architecture

- Custom CNN trained from scratch, without ImageNet or VGGFace weights.
- 4 convolutional blocks: 32, 64, 128, 256 channels.
- Depthwise separable convolutions to reduce complexity.
- Residual connections and SE attention in deeper blocks.
- Global average pooling and dropout-based classification head.

Training

- Optimizer: AdamW.
- Scheduler: cosine annealing.
- Augmentations: flip, rotation, translation, brightness, contrast, random erasing.
- Early stopping patience: 15.

Why this choice?

The model is expressive enough for FER, while still compatible with Grad-CAM and local perturbation analysis.

XAI pipeline

Grad-CAM

- Gradient-based method.
- Uses the last convolutional layer.
- Produces a smooth map of discriminative regions.

LIME

- Perturbation-based method.
- Masks image regions.
- Learns a local interpretable approximation.

SHAP

- Coalition-based attribution.
- KernelSHAP approximation on a grid.
- More expensive and more sensitive in our setup.

Common rule

For each image, the three methods explain the same target: the class predicted by the CNN.

How do we evaluate an explanation?

1. Faithfulness

- Deletion: remove important pixels first.
- Insertion: add important pixels first.
- Good explanation: low deletion, high insertion.

2. Robustness

- Small noise.
- Brightness variation.
- Small translation.
- Compare maps before and after perturbation.

3. Facial masking

- Mask the eyes.
- Mask the mouth.
- Mask the background.
- Measure confidence drop and prediction changes.

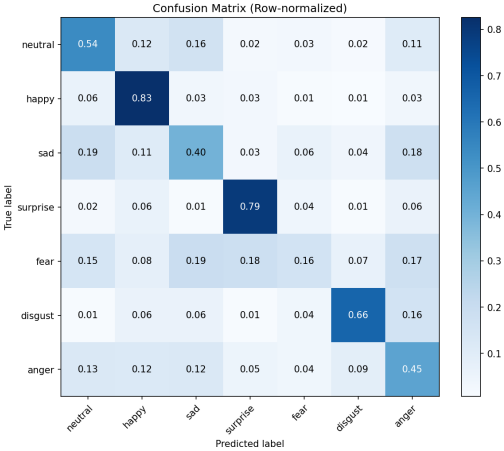
Classification performance

Global metrics on the test set

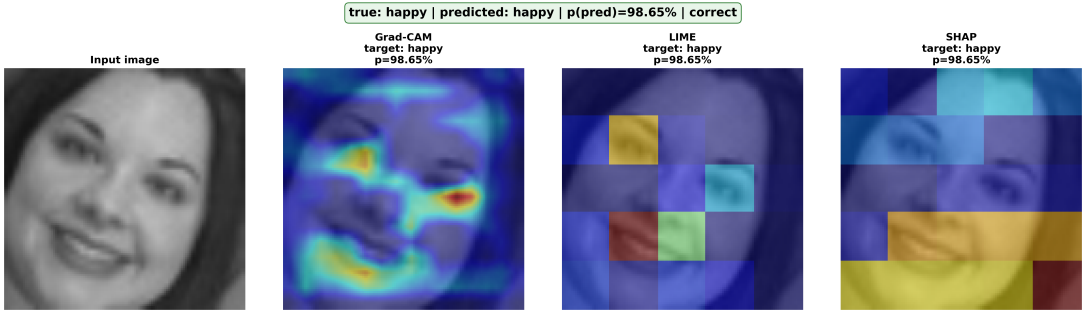
Accuracy	**55.11%**
Balanced accuracy	**54.65%**
Macro-F1	**49.11%**
Weighted-F1	**53.37%**

Reading

- The model is informative, but imperfect.
- Errors are useful to study explanation behavior.
- Strongest classes: happy and surprise.
- Most difficult class: fear.



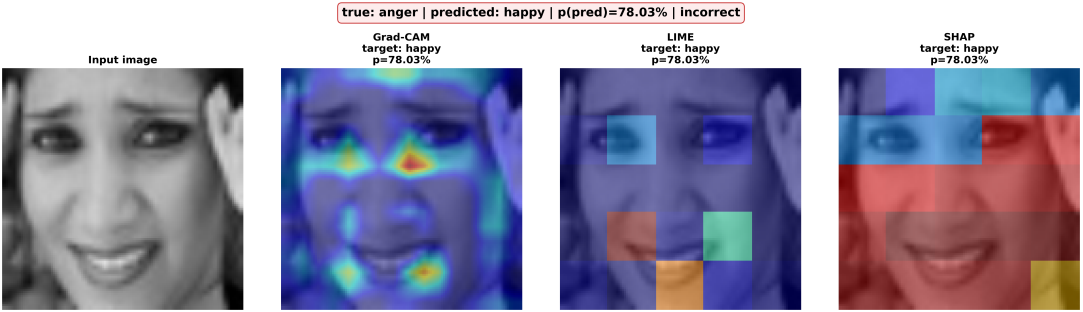
Qualitative correct case: happy → happy



Interpretation

The three methods explain the class **happy**. Grad-CAM is smoother, while LIME and SHAP highlight more discrete regions. The focus on the mouth and expressive facial traits is plausible for this prediction.

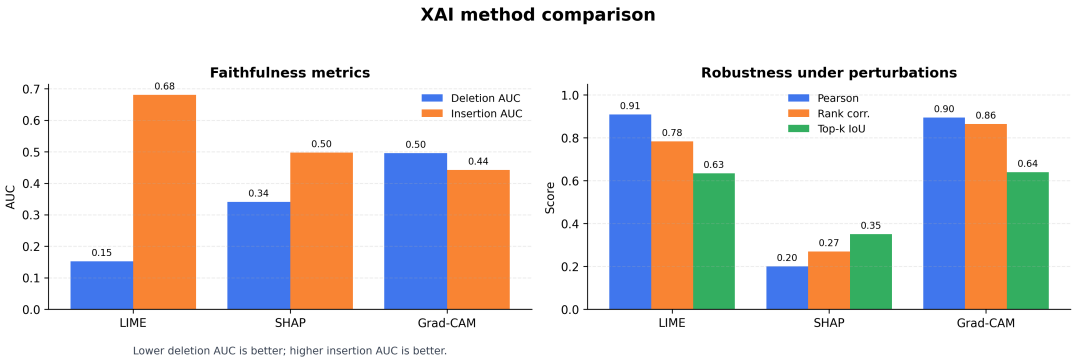
Qualitative failure case: anger → happy



Why this image matters

The true label is **anger**, but the model predicts **happy**. The explanations show that the model relies on regions compatible with a smile; the error is not random, it reveals a semantic confusion.

Quantitative comparison of XAI methods



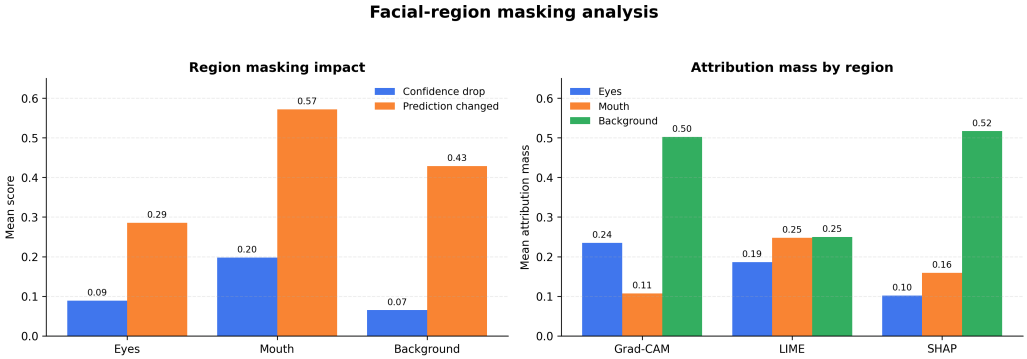
Faithfulness

LIME has the best combination: lower deletion AUC and higher insertion AUC.

Robustness

Grad-CAM and LIME remain stable; SHAP is more sensitive to perturbations in our approximation.

Facial-region masking analysis



Main result

Masking the **mouth** causes the largest average confidence drop and changes the prediction in 57% of analyzed cases. This is consistent with facial expression recognition, but it also shows that the model may over-rely on specific local cues.

Discussion: what do we actually learn?

Strengths

- End-to-end pipeline: training, evaluation, XAI.
- Comparison of three families of methods.
- Visual and quantitative evaluation.
- Error analysis, not only successful cases.

Messages

- LIME is most faithful on the analyzed sample.
- Grad-CAM is stable and readable.
- SHAP is less robust here with the grid approximation.

Limitations

- Full XAI analysis computed on 7 images due to cost.
- LIME/SHAP regions are coarse because they rely on a grid.
- The FER model is not optimized for state-of-the-art accuracy.
- Eye/mouth/background masks remain approximate.

Conclusion and take-home messages

Answer to the research question

XAI methods are not interchangeable: on our FER model, they produce visually and quantitatively different explanations.

1. Fidelity

LIME is the most convincing method in deletion/insertion tests on our sample.

2. Stability

Grad-CAM is more stable and easier to interpret, but less faithful quantitatively.

3. Failure analysis

Model errors become interpretable: some confusions arise from ambiguous facial cues.

Future work

Extend the XAI analysis to more images, use more anatomical superpixels, and train a variant with explanation-guided regularization.

Backup: per-class details

Class	Precision	Recall	F1	Support
neutral	0.512	0.541	0.526	930
happy	0.733	0.826	0.777	1349
sad	0.447	0.399	0.422	912
surprise	0.646	0.792	0.712	600
fear	0.438	0.158	0.232	768
disgust	0.230	0.659	0.341	82
anger	0.409	0.452	0.430	743

How to use this slide

If asked why the global score is not higher: RAF-DB is imbalanced and some emotions are naturally confused, especially fear, sad, surprise and anger.

Backup: files and reproducibility

Main commands

```
py -3 -m src.main_evaluate_all -config configs/main_cnn_v1.yaml -checkpoint
experiments/checkpoints/best_model150epochs.pt
```

```
py -3 -m src.main_compare_xai -config configs/main_cnn_xai.yaml -checkpoint
experiments/checkpoints/best_model150epochs.pt -output_dir experiments/xai_clean
```

Figures used

`experiments/xai_clean/report_figures/`

Quantitative outputs

`xai_method_summary.csv`, `xai_masking_summary.csv`, `test_overall_metrics.csv`

Backup: peer-reviewed references

- S. Li, W. Deng, and J. Du. Reliable Crowdsourcing and Deep Locality-Preserving Learning for Expression Recognition in the Wild. CVPR, 2017.
- R. R. Selvaraju et al. Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. ICCV, 2017.
- M. T. Ribeiro, S. Singh, and C. Guestrin. Why Should I Trust You? Explaining the Predictions of Any Classifier. KDD, 2016.
- S. M. Lundberg and S.-I. Lee. A Unified Approach to Interpreting Model Predictions. NeurIPS, 2017.
- J. Adebayo et al. Sanity Checks for Saliency Maps. NeurIPS, 2018.
- C.-K. Yeh et al. On the (In)fidelity and Sensitivity of Explanations. NeurIPS, 2019.

Credibility check

These references are standard XAI / FER papers from CVPR, ICCV, KDD and NeurIPS, and directly justify the dataset, methods and evaluation protocol.